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STRATEGIC PAPERS

Systemic Fragility

and

Strategic Necessity

A Comprehensive Life-Cycle Assessment of the

F-35 Lightning II Program

and Western Defense Industrial Capacity
in an Era of Great Power Competition

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KEY FINDINGS AT A GLANCE

$2+ TRILLION Program Lifecycle Cost	**50%** Fleet Availability Rate
232:1 China-US Shipbuilding Capacity	**3-4x** Russian Artillery Production Advantage
$100+ BILLION Cancelled Defense Programs	**238 DAYS** Average F-35 Delivery Delay (2024)

STRATEGIC WARNING

War games consistently demonstrate that U.S. forces would exhaust long-range precision munitions within days of a major conflict. The defense industrial base cannot surge production quickly enough to matter in plausible warfighting timelines.

EXECUTIVE SUMMARY

The F-35 Lightning II Joint Strike Fighter stands as the most expensive weapon system in human history, with lifecycle costs now exceeding **$2 trillion** through 2088. This assessment examines whether the program's documented performance deficiencies—50% fleet availability rates, persistent cost escalation, chronic schedule delays, and critical supply chain vulnerabilities—can be reconciled with its strategic necessity in an era of great power competition where adversaries are rapidly closing capability gaps while massively outproducing Western defense industries.

The strategic context is unprecedented. Russia now manufactures approximately 4.5 million artillery shells annually—nearly four times NATO's combined production. China has achieved 232 times greater shipbuilding capacity than the United States while producing 100-120 fifth-generation fighters annually. War games consistently demonstrate that U.S. forces would exhaust long-range precision munitions within days of a major conflict, and the defense industrial base cannot surge production quickly enough to matter in plausible warfighting timelines.

This analysis finds that the F-35 program epitomizes systemic dysfunction in Western defense acquisition. The practice of 'concurrency'—beginning mass production before completing development—has created a perpetual retrofit burden. Block 4 modernization is $6 billion over budget and five years behind schedule. Technology Refresh 3 delays caused a year-long delivery halt, and all 110 aircraft delivered in 2024 arrived late, averaging 238 days behind schedule. Meanwhile, China flew two sixth-generation fighter prototypes in December 2024—years ahead of any U.S. demonstration.

The supply chain presents existential vulnerabilities. Each F-35 contains over 400 kilograms of rare earth materials with critical dependencies on Chinese sources.

China controls 70% of global rare earth extraction and over 90% of downstream processing—including 100% monopoly on separation capacity for dysprosium and terbium essential for the aircraft's advanced electronics. Production stoppages from discovered Chinese-sourced components occurred in 2022, 2023, and 2024.

The broader pattern of Western defense procurement failure compounds these concerns. Over $100 billion in cancelled programs since 1999—Future Combat Systems, Comanche, Crusader, GCV, Zumwalt, LCS, Constellation—consumed resources without delivering needed capabilities. The Army has not fielded a new manned ground combat vehicle since the M2 Bradley in the early 1980s despite four successive replacement attempts.

Strategic flexibility is not weakness—it is wisdom in the face of documented evidence that original assumptions have proven catastrophically wrong.

1. THE TWO-TRILLION DOLLAR ENTERPRISE

The F-35 Lightning II Joint Strike Fighter (JSF) stands as the singular most expensive and industrially complex weapon system in the history of warfare. Conceived in the post-Cold War era as a universal solution to tactical aviation—a 'Swiss Army knife' designed to replace the A-10 Thunderbolt II, F-16 Fighting Falcon, F/A-18 Hornet, and AV-8B Harrier II—the program has metastasized into a geopolitical and fiscal colossus. The Department of Defense (DOD) now estimates the total life-cycle cost of the program to exceed $2 trillion, a figure that encompasses research, development, procurement, and decades of operations and sustainment through the year 2088.

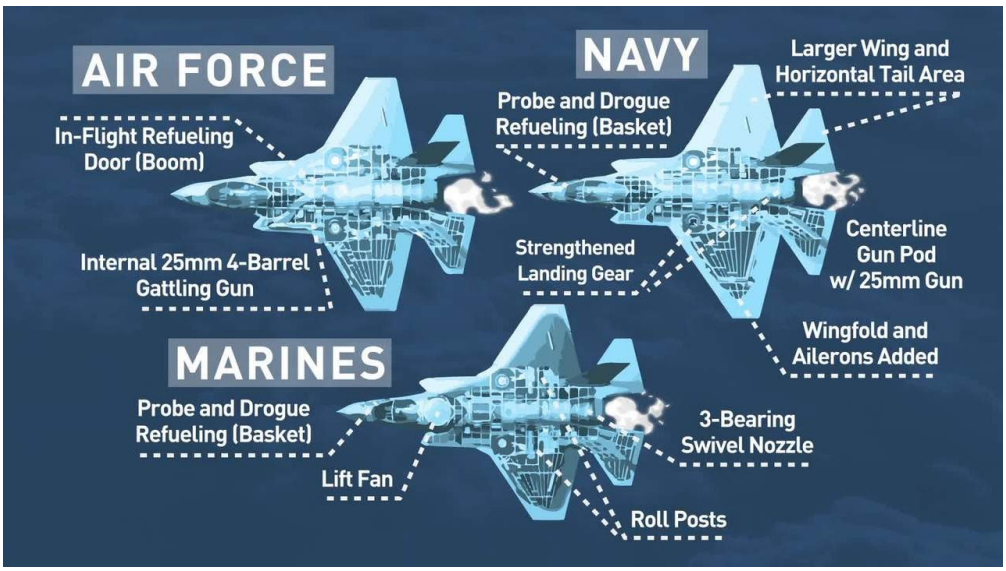

Figure 1: The three F-35 variants—Air Force, Navy, and Marines—each with unique requirements that undermined commonality goals.

This staggering valuation has made the F-35 a lightning rod for criticism, embodying the tension between technological ambition and fiscal reality. To its proponents, the F-35 is the 'quarterback of the sky,' a stealthy sensor node capable of penetrating the world's most dense Integrated Air Defense Systems (IADS) and distributing critical targeting data across the joint force. To its detractors, it is the

ultimate manifestation of the military-industrial complex's excesses—an 'infinite money pit' that produces a fragile, high-maintenance asset plagued by software instability and operational limitations.

The program's scale is difficult to overstate. It involves a global industrial base with final assembly and checkout (FACO) facilities in the United States, Italy, and Japan, and a supply chain that spans nearly every NATO member state. Yet, this very interdependence, designed to lower costs through economies of scale and cement diplomatic alliances, has introduced critical vulnerabilities. The program is currently navigating a precarious 'Block 4' modernization phase, struggling with the integration of the Technology Refresh 3 (TR-3) hardware, and facing profound supply chain threats from strategic competitors who control the raw materials essential for the aircraft's advanced avionics and propulsion systems.

As of 2025, the program has delivered over 1,000 aircraft, yet it remains in a state of perpetual development. The operational reality of the F-35 is defined by a dichotomy: it is simultaneously the most advanced fighter in the world, achieving high export success and integration into allied air forces, and a platform that struggles to maintain mission-capable rates above 55%, leaving nearly half the fleet grounded or restricted in capability at any given time.

2. FINANCIAL ARCHITECTURE: THE CONCURRENCY TAX

The epithet 'infinite money pit' is derived from a pattern of continuous cost growth and schedule slippage that has defined the JSF program since its inception. The financial architecture of the F-35 is bifurcated into acquisition costs (buying the jets) and sustainment costs (flying and fixing them), both of which have exhibited inflationary behaviors driven by structural decisions made early in the program's history.

2.1 Concurrency: Acquisition Malpractice

A foundational error in the F-35's acquisition strategy was the adoption of 'concurrency'—the policy of beginning mass production while development and testing were still ongoing. The rationale was to accelerate fielding and avoid gaps in fighter inventory. In practice, this created a 'concurrency tax.' Hundreds of aircraft were produced with immature hardware and software configurations, requiring expensive post-production retrofits to meet the standards established by later testing.

Under Secretary Frank Kendall characterized concurrency as ***'acquisition malpractice'*** in 2014. The F-35 program built over 600 aircraft before completing Initial Operational Test and Evaluation, requiring an estimated $1.4-1.7 billion in retrofits for already-produced aircraft. The DOD purchased one-third of all planned F-35s before validating that the aircraft met requirements.

The financial impact of this strategy is evident in the escalation of acquisition costs. The program's total acquisition cost (development plus procurement) has risen from a 2012 baseline of $395.7 billion to a 2023 estimate of $485.2 billion, an

increase of nearly $90 billion. This growth is not merely a function of buying more aircraft but reflects the capital-intensive nature of correcting design deficiencies discovered after metal was already cut.

2.2 Sustainment Cost Explosion

Sustainment costs present perhaps the most concerning long-term burden. The GAO's April 2024 assessment documented total lifecycle costs of $2.063 trillion through 2088—nearly ten times the original estimates. Sustainment costs alone increased 44% in just five years (2018-2023), from $1.1 trillion to $1.58 trillion, despite the fleet flying fewer hours than originally planned.

COST PER FLIGHT HOUR COMPARISON

F-35A: $42,000 vs F-16: $26,000

60% higher than the aircraft it was designed to replace

The Air Force abandoned its original affordability target of $4.1 million per tail per year, raising it to $6.8 million (a 66% increase) simply to claim the target was 'met.' Services are now meeting cost targets by planning to fly aircraft 21% fewer hours than originally projected. Despite these documented failures, Lockheed Martin received $1.7 billion in sustainment payouts without financial penalties because the Joint Program Office failed to include readiness metrics in contracts.

3. OPERATIONAL REALITY: A FLEET THAT CANNOT FLY

The most damning indictment of the F-35 program comes not from critics but from the Pentagon's own oversight bodies. The December 2024 DOD Inspector General report documented that F-35 fleet availability averaged just 50% in FY2024—meaning aircraft were unavailable to fly half the time. This rate fell 17 percentage points below the Pentagon's minimum performance requirement.

Figure 2: F-35 incident documentation—the human and material cost of operational failures.

3.1 Mission Capability in Decline

The trajectory is moving in the wrong direction. Mission-capable rates declined from 69% in 2021 to 51.5% in 2024, while Full Mission Capable rates—the aircraft's ability to perform all assigned combat roles—hover around just 30%. No F-35 variant has met its desired mission-capable rate in six consecutive fiscal years (2018-2023) according to GAO assessments.

READINESS METRICS: THE DECLINE

Mission-Capable Rate: 69% (2021) → 51.5% (2024)

Full Mission Capable Rate: ~30%

Six consecutive years failing to meet readiness targets

3.2 Block 4 Modernization Crisis

The Block 4 modernization program, intended to deliver 66 new capabilities by 2026, is now at least $6 billion over budget and five years behind schedule, with completion pushed to 2031 at earliest. The Technology Refresh 3 (TR-3) upgrade—a $1.9 billion hardware/software suite critical to combat capability—caused a year-long delivery halt from July 2023 to July 2024. When deliveries resumed, aircraft arrived with 'truncated' software capable of training only, not combat operations.

All 110 F-35s delivered in 2024 were late, averaging 238 days behind schedule—up from 16 days in 2021. Engine production presents another critical vulnerability: all 123 engines delivered in 2024 arrived late, averaging 238 days behind schedule (up from 68 days in 2023). Root causes include parts arriving late from suppliers, defective parts requiring rework, and assembled engines failing testing standards.

4. OPERATIONAL INCIDENTS: THE OCEAN CLAIMS ANOTHER

Beyond readiness statistics, the F-35 program has been marked by a series of high-profile operational incidents that underscore the aircraft's fragility and the strategic security implications of losing $100+ million platforms in operationally significant locations.

Figure 3: January 2022 South China Sea incident—$100M F-35C lost overboard from USS Carl Vinson, creating intelligence recovery race with China.

4.1 The South China Sea Incident

On January 24, 2022, an F-35C Lightning II crashed during landing operations aboard the aircraft carrier USS Carl Vinson in the South China Sea. The pilot ejected safely but seven sailors were injured. The $100 million aircraft—carrying some of America's most closely guarded stealth technology—sank to a depth of approximately 12,400 feet in waters contested by China.

What followed was a race against time and against Chinese intelligence services to recover the aircraft before adversaries could access its classified systems. The Navy deployed the deep-sea salvage vessel USNS Henson with specialized equipment to locate and recover the jet, successfully raising it on March 2, 2022—37 days after the crash. The operation's urgency highlighted a fundamental vulnerability: each F-35 loss in contested waters creates a potential intelligence windfall for adversaries.

Figure 4: F-35 floating in the South China Sea following the carrier landing incident—stealth technology exposed to potential adversary recovery.

4.2 Pattern of Ground Incidents

Figure 5: Multiple F-35 incidents have been captured on camera, documenting the operational fragility of America's most advanced fighter.

The F-35 has experienced multiple Class A mishaps (incidents causing death, permanent disability, or damages exceeding $2.5 million). Beyond the South China Sea incident, notable losses include crashes in Japan, South Carolina, Texas, and California. Each incident consumes not merely the $80-100 million airframe cost but extensive investigation resources, potential loss of trained pilots, and erosion of operational readiness.

5. SUPPLY CHAIN VULNERABILITIES: DEPENDENT ON THE ADVERSARY

The F-35's globalized supply chain, once touted as a strength through allied industrial participation, has revealed critical vulnerabilities that pose existential risks to program viability. The aircraft designed to defeat Chinese and Russian air defenses depends on Chinese raw materials for its most critical components.

5.1 Rare Earth Dependencies

Each F-35 contains over **400 kilograms of rare earth materials**, with critical components dependent on Chinese sources. China controls 70% of global rare earth extraction and over 90% of downstream processing—including **100% monopoly** on separation capacity for dysprosium and terbium, essential for permanent magnets capable of withstanding high temperatures in the F135 engine and avionics systems.

CRITICAL STRATEGIC VULNERABILITY

China controls 70% of rare earth extraction

China controls 90%+ of rare earth processing

China has 100% monopoly on dysprosium/terbium separation

Pentagon projects 10-20 years to achieve meaningful independence

These dependencies are not theoretical concerns. In September 2022, F-35 deliveries were suspended after discovering Chinese-sourced cobalt-samarium alloy magnets in Honeywell turbomachine components, violating U.S. procurement law. Similar discoveries caused six-month production stoppages in both 2023 and 2024. China's April 2025 export controls on seven critical minerals and bans on rare earth processing technology exports weaponize this advantage.

5.2 Single-Source Vulnerabilities

Turkey's 2019 removal from the program after accepting Russian S-400 systems demonstrated how quickly supply chain assumptions can unravel. Turkish companies manufactured 937 F-35 parts, with 400 sole-sourced to Turkish firms—no alternative suppliers existed. Retooling the industrial base cost $500-600 million in non-recurring engineering, with projections of 50-75 aircraft delays over two years.

Pratt & Whitney's F135 remains the sole powerplant, creating strategic risk that no alternative exists if production falters. As of February 2025, more than 4,000 parts shortages existed in final production stage—twice the historic average. The GAO documented a parts backlog exceeding 10,000 items awaiting repair, with average repair times of 141 days versus the 60-day target. Depot maintenance facilities remain 12 years behind schedule in establishing required capabilities.

6. THE CLOSING WINDOW: COMPETITOR CAPABILITIES

The strategic premise underlying F-35 acquisition—maintaining decisive advantage over adversary aircraft—requires examination against current production realities and the emergence of sixth-generation competitors.

6.1 China's Accelerating Air Power

China's J-20 program has achieved production rates that fundamentally alter regional air power balance. CSIS documented 230 fifth-generation fighters in PLA service by 2024, with the J-20 fleet now exceeding 300 aircraft and growing at 70-100 per year. This production rate means China's J-20 fleet has already surpassed total F-22 production (187 aircraft) and could approach 800-1,000 by 2030 at current trajectories.

CHINA'S SIXTH-GENERATION LEAD

December 26, 2024: China reveals two 6th-gen prototypes (J-36, J-50)

December 2025: Three J-36 prototypes flying

2028: U.S. F-47 (NGAD) first flight expected

China is years ahead in sixth-generation development

More concerningly, China revealed two distinct sixth-generation prototypes on December 26, 2024—the Chengdu J-36 trijet flying-wing and the Shenyang J-50—years ahead of any U.S. flight demonstrations. Three J-36 prototypes were flying by December 2025, while the Boeing F-47 (U.S. NGAD) first flight isn't expected until 2028. This development timeline suggests the F-35 may face adversary aircraft specifically designed to counter its capabilities before many allied fleets achieve full operational capability.

7. WESTERN DEFENSE INDUSTRIAL DECLINE

The F-35's problems exist within a broader pattern of Western defense procurement dysfunction that has consumed over $100 billion in cancelled programs while failing to deliver needed capabilities.

7.1 The $30 Billion Ground Vehicle Graveyard

Since 1999, the Army has spent approximately $30 billion on cancelled vehicle programs while failing to field a single new manned ground combat vehicle. The Army still relies on the M2 Bradley—a design from the early 1980s—because four successive replacement attempts have collapsed:

- Future Combat Systems (FCS): $18-20 billion spent, cancelled 2009
- RAH-66 Comanche helicopter: $6.9 billion spent, cancelled 2004
- XM2001 Crusader artillery: $2 billion spent, cancelled 2002
- Ground Combat Vehicle (GCV): $1 billion spent, cancelled 2014

7.2 Naval Procurement Dysfunction

The Zumwalt-class destroyer stands as perhaps the most dramatic example of requirements collapse. Originally planned as 32 ships, the program was reduced to just three—driving per-ship costs from an initial $3.3 billion estimate to $7.5 billion including R&D ($22.5 billion total program cost for three vessels). The ships' ammunition costs escalated from a $35,000 target in 2004 to $800,000-1,000,000 per round by 2016—forcing cancellation of the ammunition program and leaving the destroyers commissioned without functioning main guns.

The Littoral Combat Ship program has consumed an estimated $100 billion lifetime cost while delivering vessels so problematic that the Navy began decommissioning them after as little as five years of service. The Constellation-class frigate program was cancelled in November 2025 after the lead ship reached only 10% completion with a 36-month schedule slip.

7.3 The Hypersonic Gap

The Army's Dark Eagle Long-Range Hypersonic Weapon program exemplifies broader procurement dysfunction. After spending $2.69 billion on the first battery alone and experiencing multiple test failures, Dark Eagle achieved Initial Operating Capability in September 2025—roughly six years after China fielded the DF-17 and Russia deployed the Avangard hypersonic glide vehicle. Per-missile costs have exceeded $41 million, and the Army Chief of Staff publicly acknowledged that alternative missiles at 'a tenth of the price' might provide better value.

8. THE PRODUCTION CAPACITY GAP

Three years of supporting Ukraine have exposed the fragility of Western defense industrial capacity—a reality with direct implications for F-35 sustainment and broader deterrence credibility.

8.1 Russia's Wartime Industrial Transformation

Russia's defense industrial base has demonstrated a capacity for wartime mobilization that Western planners failed to anticipate. Artillery shell production has increased approximately 11-fold since 2022, from roughly 400,000 rounds to an estimated 4-4.5 million annually. NATO Secretary General Rutte confirmed Russia produces approximately four times greater artillery output than the entire NATO alliance.

RUSSIA	NATO COMBINED
4-4.5 Million	**1.7-2 Million**
Artillery Shells/Year	Artillery Shells/Year

8.2 China's Industrial Superiority

Chinese defense industrial production has achieved quantitative advantages that fundamentally alter strategic calculations. The Office of Naval Intelligence's assessment that Chinese shipbuilding capacity exceeds U.S. capacity by 232:1 represents a transformative reality. China represents approximately 50% of global shipbuilding capacity; one Chinese shipyard has more capacity than all U.S. shipyards combined.

China has built 7-8 Type 093B nuclear attack submarines in the past three years—exceeding the number of attack submarines China constructed over the past three decades. Fighter aircraft production shows comparable acceleration, with J-20 production reaching an estimated 100-120 aircraft annually.

9. WAR GAME IMPLICATIONS: EXHAUSTION IN DAYS

CSIS Taiwan invasion war games—24 iterations across 2022-2023—found that U.S. forces would deplete Long-Range Anti-Ship Missiles within the first week in all scenarios. The games projected expenditure of 5,000+ long-range missiles in three weeks, including approximately 4,000 JASSM missiles, 450 LRASMs, 400 Harpoons, and 400 Tomahawks.

WAR GAME FINDINGS

CSIS Taiwan Scenarios (24 iterations):

Long-Range Anti-Ship Missiles depleted: Week 1

5,000+ long-range missiles expended: 3 weeks

Average time to replace inventory: 8.4 years

"China is acquiring high-end weapons systems 5-6 times faster than the United States."
— CSIS Assessment

The 2024 House Select Committee on the CCP simulation found U.S. and Taiwan running low on critical munitions 'in a matter of days.' British exercises produced comparable findings: the UK Warfighter 21-4 exercise saw the 3rd Division exhaust national stockpiles of critical munitions in just over one week.

These consumption rates cannot be replaced with current production capacity. CSIS analysis found an average of 8.4 years to replace major weapon system inventories at surge production rates. Factory investment implementation requires a minimum 18-24 months. General Mark Milley stated that in 'a great power war between the United States and Russia or the United States and China, the consumption rates [of conventional munitions] would be off the charts.'

10. CASE STUDY: CANADA'S F-35 DECISION

Canada's pending C$73.9 billion F-35 commitment (88 aircraft over 45 years) illustrates the strategic calculus facing allied nations. The case merits examination as a model for evaluating F-35 acquisition decisions.

10.1 Cost Escalation Pattern

The Canadian acquisition cost has risen from an original C$19 billion to C$27.7 billion—a 46% increase in just two years. The Parliamentary Budget Officer's November 2023 analysis estimated C$73.9 billion in total lifecycle costs, with operations and sustainment comprising C$53.8 billion—the overwhelming majority of program expense. These figures translate to approximately C$840 million per aircraft over the operational lifetime.

CANADIAN F-35 COST ESCALATION

C$19 Billion → **C$27.7 Billion**

46% increase in two years

Total Lifecycle: C$73.9 Billion (C$840M per aircraft)

10.2 The Mixed Fleet Alternative

The Saab Gripen E presents an alternative worth serious consideration. Operating costs are claimed at $4,700-$10,000 per flight hour versus the F-35's $42,000—potentially 60% lower lifecycle costs. The Gripen was designed for Arctic operations with demonstrated capability at -20°C to -26°C, operating from 800-meter dispersed road bases versus F-35's requirement for prepared runways.

The Australian experience offers a model: the Royal Australian Air Force successfully operates a mixed fleet of 72 F-35As alongside 24 Super Hornets and 12 EA-18G Growlers. A Canadian mixed fleet—the 16 contractually committed F-35s plus alternatives for sovereignty and patrol missions—would preserve NATO interoperability where essential while reducing fiscal exposure and establishing industrial sovereignty.

11. CONCLUSIONS AND STRATEGIC RECOMMENDATIONS

This assessment reaches three principal conclusions with significant implications for defense planning and procurement decisions.

11.1 The F-35 Cannot Be the Sole Solution

The F-35's documented performance deficiencies—50% fleet availability, persistent cost escalation, supply chain vulnerabilities, and schedule delays—preclude reliance on it as the universal solution originally envisioned. The platform may remain strategically necessary for high-end penetrating strike missions in contested environments, but its operational limitations, maintenance burden, and cost structure make it unsuitable as the exclusive tactical aviation asset for any nation.

11.2 Production Capacity Is Deterrence

The combination of procurement dysfunction, production capacity gaps, and adversary industrial mobilization creates strategic vulnerabilities that existing planning frameworks may not adequately address. The defense industrial base is now a strategic asset equivalent to deployed forces—yet has been managed for efficiency rather than resilience, for profit rather than capacity, and for peacetime rather than wartime requirements.

11.3 Strategic Flexibility Is Essential

Nations retain flexibility to pursue mixed-fleet approaches, invest in domestic defense industrial capacity, and evaluate alternatives rather than committing exclusively to troubled programs. The prudent course is exercising this flexibility—committing to genuine competitive evaluation of alternatives, examining proven mixed-fleet models, and ensuring that defense procurement serves national interests rather than institutional momentum.

11.4 Specific Recommendations

- Conduct rigorous independent lifecycle cost analysis before committing to F-35 acquisition, using conservative assumptions derived from documented program performance rather than contractor projections.
- Evaluate mixed-fleet alternatives that preserve interoperability for coalition high-end missions while providing cost-effective platforms for sovereignty, patrol, and training missions.
- Prioritize domestic defense industrial capacity investment over imported turnkey solutions, recognizing that supply chain resilience is itself a strategic capability.
- Establish performance-based contract structures with meaningful readiness metrics and financial penalties for non-compliance.
- Invest in munitions production capacity and strategic stockpiles, recognizing that sophisticated platforms are of limited utility without adequate weapons inventories.
- Develop contingency plans for rare earth supply chain disruption, including accelerated domestic processing capability development.

FINAL ASSESSMENT

*Strategic flexibility is not weakness—it is wisdom in the face of documented evidence that original assumptions have proven catastrophically wrong. The F-35 may remain a necessary component of allied air power, but it cannot be the **only** component. Production capacity is deterrence; the West's current capacity may be insufficient to deter risk-tolerant adversaries who have demonstrated*

their own industrial mobilization capabilities.

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